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pygame 2.5.1 (SDL 2.28.2, Python 3.9.6)
Hello from the pygame community. https://www.pygame.org/contribute.html
WARNING:root:Tried to import the tfLite runtime, but it was not found. Trying to switching to onnxruntime instead, if appropriate
available.
🔧 Advanced Alexa with OpenCV is ready! Say 'alexa'

🔥 Wake word detected!
Alexa: Yes?
Listening for command...
You said: what do you see
Alexa: Taking a photo...
📷 Photo captured successfully
Alexa: Analyzing the scene...
Vision Error: name 'base64' is not defined
Alexa: Sorry, I couldn't analyze the image.

🔥 Wake word detected!
Alexa: Yes?
Listening for command...
You said: translate my language tu english
Alexa: Okay, I can definitely help you with that!

Please **paste or type the text** you want translated, and I will translate it from your language to English.

If you know what language it is, feel free to tell me, but I can often detect it myself!

🔥 Wake word detected!
Alexa: Yes?
Listening for command...
You said: tum kya kar rahe ho isko angreji mein translate karke batao
Alexa: "तुम क्या कर रहे हो?" का अंग्रेजी अनुवाद है:

```

Fig. 11: Raspberry conversation log

**TO DOWNLOAD SOURCE CODE
AND RESOURCES**



Scan this QR code or go to
<https://sourcecode.electronicshobby.com>
to access and download source code
and other resources for this project

Using the generative AI features

To enable the generative AI capabilities, switch the audio input source back to the microphone connected through the voice bonnet. Voice commands can then be used to interact with the speaker. Example commands include:

- Tell me today's weather report.
- What are the latest news

updates?

- What do you think about the current growth of the electronics industry?
- Create a short story for children.
- Tell me a joke.

The speaker processes the spoken request, sends it to the configured large language model (LLM), and generates detailed, conversational, and context-aware responses.

Using the visual AI features

The system also supports multimodal AI capabilities through its integrated camera. For example, the command "What do you see?" prompts the speaker to capture an image and describe the scene in front of it.

Similarly, a book, document, or printed page can be placed in front of the camera, followed by

the command "Read this page for me." The device then performs optical character recognition (OCR), extracts the text, and reads it aloud.

In addition, the speaker can function as a translator, interpreter, and transcription device. It can translate text or speech into different languages, automatically detect the source language, and provide translated responses in real time.

These multimodal AI capabilities make the device significantly more powerful than a conventional smart speaker, enabling a wide range of intelligent voice- and vision-based interactions. Fig. 11 shows a sample conversation log captured during system testing. **EFY**

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